

Permitless Concealed Carry Adoption and Firearm Mortality in U.S. States, 1999–2024: A Source-Audited Panel Analysis

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Abstract

Permitless concealed carry laws remove the requirement that eligible adults obtain a carry permit before carrying a concealed handgun in public. These laws spread rapidly during the last decade, but the empirical question is difficult because recent adopters are politically and structurally patterned, legal coding contains edge cases, and permitless carry often changes a bundle of pre-carry screens rather than a single mechanism. We assembled a 50-state annual panel from 1999 through 2024 using CDC mortality outcomes and a source-audited permitless-carry treatment file that records clean adopters, non-adopters, baseline-permitless states, partial cases, and ambiguous cases. Firearm suicide was the primary outcome. In the baseline state and year fixed-effects model, adjusted for unemployment and per-capita income with state-clustered standard errors, permitless carry adoption was associated with a 1.391 deaths per 100,000 increase in firearm-suicide rates. Age-adjusted firearm-suicide models gave a similar estimate (1.407). The association remained positive in population-weighted models, COVID-period exclusions, external firearm-law covariate models, leave-one-adopter-out checks, legal edge-case sensitivity analyses, and cohort-window ATT estimates, although the cohort-window estimates were smaller and state-specific trends attenuated the estimate. Total suicide also increased in the baseline model, while non-firearm suicide was positive but less stable. Firearm homicide showed no comparable association, although homicide-rate missingness was concentrated in small-population state-years. These findings support a state-level observational association between permitless concealed carry adoption and firearm suicide.

Introduction

Permitless concealed carry is a recent policy margin inside a much older concealed-carry debate. Earlier right-to-carry studies focused mostly on transitions from no-issue or may-issue systems to shall-issue permitting, asking whether broader legal carrying deters violent crime or increases firearm violence [1–3]. That debate remains methodologically contested, and replication work has emphasized how sensitive concealed-carry estimates can be to event-study validity, heterogeneous treatment effects, and influential observations [4]. The policy landscape has also moved again. Since 2015, many states have gone beyond shall-issue permitting by allowing eligible adults to carry concealed handguns without first obtaining a carry permit, a shift that became even more salient after *Bruen* changed the constitutional environment for restrictive carry permitting [5]. The

empirical problem is therefore no longer only whether permits should be granted more easily; it is what happens when the permit step itself is removed.

That distinction matters because the permit step can screen, delay, train, or document the person who wants to carry. A permitless-carry law may remove a training requirement, a permit-background-check requirement, a permit-specific misdemeanor screen, or a discretionary suitability review. It may also leave substantive firearm-possession prohibitions in place while eliminating the pre-carry application. Prior concealed-carry work has emphasized that these permitting provisions can matter for violence outcomes [2]. A recent permitless-carry working paper estimates increases in violent crime and firearm aggravated assault after adoption, reinforcing the need to evaluate the newer permitless margin separately from older shall-issue transitions [6]. Firearm-suicide work has also reported that concealed-carry permit requirements are associated with lower firearm suicide rates, alongside handgun purchase permits and waiting periods [7]. The policy margin is therefore plausible for suicide as well as homicide, but the relevant mechanism is not self-evident.

The suicide question deserves separate treatment from the homicide question. Firearm access is strongly associated with firearm suicide at household, individual, and state levels [8–10]. Purchaser-licensing evidence also links stronger handgun screening systems with lower firearm homicide and suicide in several state settings [11]. A law that changes public carrying may not directly change household gun ownership, but it can alter the carrying environment, the perceived normality of immediate firearm availability, and the screening requirements attached to carrying a handgun. Those pathways are closer to lethal-means access than to the deterrence theory that dominates older right-to-carry crime debates. For that reason, a permitless-carry mortality study should not treat homicide and suicide as interchangeable endpoints.

The main empirical risk is that permitless carry adoption is not random. States that adopted permitless carry in this period tended to have higher baseline gun ownership, higher rurality, higher Republican presidential vote share, and higher baseline firearm-suicide rates than states that did not adopt. These differences do not invalidate a panel design, but they do raise the standard for interpretation. A simple post-adoption coefficient can mix the policy association with pre-existing state trajectories, other firearm laws, political selection, health-access differences, substance-use environments, poverty, age structure, and mental-health workforce access.

Legal coding creates a second risk. Permitless open carry is not the same as permitless concealed carry, and several states do not fit cleanly into a binary adoption map. Vermont entered the panel already permitless, so it is not a within-panel adoption event. Arkansas has competing legal interpretations before and after its 2023 statutory clarification. Mississippi’s 2015 and 2016 changes differ in how broadly permitless concealed carry applied. A paper that ignores these details can report a precise estimate attached to an imprecise treatment.

We address these risks by combining a source-audited legal treatment file with a simple results hierarchy. Firearm suicide is the primary outcome. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths are secondary checks. State-specific trends, staggered-adoption summaries, event-time checks, Arkansas recoding, leave-one-adopter-out estimates, placebo timing, and covariate-layer models are used to show where the association persists and where it weakens.

This study treats permitless concealed carry as a distinct recent policy rather than as the endpoint of older right-to-carry classifications. We ask whether states adopting permitless concealed carry experienced larger post-adoption changes in firearm suicide rates, and whether that pattern was also present for firearm homicide.

Data and Treatment Coding

State-year panel

The analysis uses an annual 50-state panel from 1999 through 2024. Mortality outcomes come from CDC WONDER underlying-cause mortality files and are expressed as crude rates per 100,000 residents [12]. Crude rates are the primary outcome scale because they measure the realized mortality burden in each state-year. Age-adjusted WONDER rates, standardized to the 2000 U.S. standard population, are used as demographic-composition sensitivity checks for firearm suicide, total suicide, firearm homicide, and total firearm deaths. CDC WISQARS documentation provides the broader injury-surveillance context for fatal injury data derived from National Center for Health Statistics mortality files [13]. The primary clean-adoption models use 1,222 state-year observations from 26 source-verified within-panel adopters and 21 verified non-adopters. Firearm-homicide models use 1,142 observations because some state-year homicide rates are unavailable or suppressed. The primary outcome is firearm suicide. Secondary outcomes are total suicide, non-firearm suicide, firearm homicide, and total firearm deaths.

The baseline state-year covariates are unemployment and per-capita income. Sensitivity layers add external firearm-law indicators from the Tufts State Firearm Law Database [14], health-insurance access from Census SAHIE, drug-overdose mortality, Census demographic and poverty measures, and HRSA mental-health-provider access. RAND state-level household firearm-ownership estimates, rurality, presidential vote share, and baseline firearm-suicide rates are used for descriptive selection and heterogeneity analyses [15].

Treatment coding

The treatment is statewide permitless concealed carry: eligible adults can carry a concealed handgun in public without first obtaining a carry permit. The annual treatment indicator turns on in the first calendar year the statewide permitless concealed-carry rule is effective. Permitless open carry alone is not treated as permitless concealed carry.

The legal audit contains one row per state. It records 26 source-verified current-adopter rows, 21 verified non-adopter rows, one partial row, one ambiguous reviewed row, and one baseline-permitless row (Table 1). The primary analysis excludes the partial, ambiguous, and baseline-permitless rows so the headline estimate compares clean within-panel adopters with verified non-adopters. Mississippi, Arkansas, and Vermont remain visible in the legal audit rather than being hidden inside the binary treatment variable. Arkansas is tested separately under 2021 and 2023 coding, Mississippi is reintroduced in edge-case sensitivity runs, and Vermont is recorded as baseline permitless rather than as a within-panel adoption event. Supplementary Note 2 summarizes the audit structure and edge-case handling; the full state-by-state source trail is maintained in the repository legal-audit file.

Table 1: **Source-audited permitless-carry treatment map.**

Audit status	State count
Source-verified current adopter	26
Verified non-adopter	21
Ambiguous reviewed	1
Partial	1
Baseline permitless	1

The primary analytic panel contains 26 clean source-verified adoption-year states and 21 verified non-adopter states. Mississippi, Arkansas, and Vermont are handled outside the primary clean-adoption estimate.

The audit also records carry-permit features for the 26 source-verified adopter rows. Twenty-one had a training requirement removed, 25 removed a carry-permit background-check screen, and 12 removed a permit-specific misdemeanor-violence screen. These fields describe the carry-permit process, not every underlying firearm-possession prohibition. We therefore treat them as descriptive policy features rather than as definitive mechanism tests.

Empirical Strategy

The central specification is a two-way fixed-effects state-year regression:

$$Y_{st} = \alpha_s + \lambda_t + \beta D_{st} + X'_{st}\theta + \varepsilon_{st},$$

Here Y_{st} is the mortality rate for state s in year t , D_{st} equals one in and after the first calendar year statewide permitless concealed carry becomes effective, α_s and λ_t are state and year fixed effects, and X_{st} contains unemployment and per-capita income in the baseline model. Standard errors are clustered by state. The coefficient β is the adjusted post-adoption association: the average within-state change after adoption, net of national year shocks and the baseline covariates. The unweighted model targets an average-state association; population-weighted models are reported separately because they answer the resident-weighted version of the same question.

This estimand is familiar in firearm-policy panel studies, but staggered adoption complicates interpretation when effects vary across cohorts or time [16–19]. We therefore report cohort-window ATT summaries, not-yet-treated dynamic ATT estimates, and never-treated-control event-time fixed-effect estimates. These analyses ask whether the direction of the result is similar under alternative timing comparisons and whether pre-adoption event-time patterns weaken a parallel-trends interpretation [20].

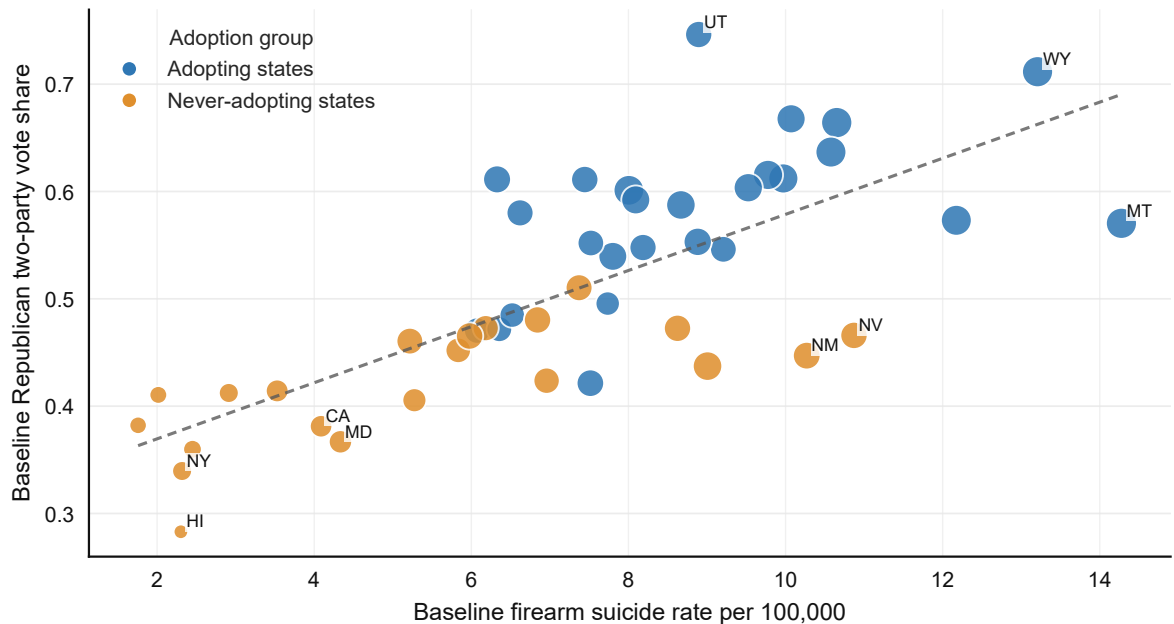
The outcome ordering was fixed before drafting the manuscript but was not preregistered. Firearm suicide is primary. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths assess outcome specificity. Sensitivity analyses test alternative covariate sets, age-adjusted rates, state-specific trends, COVID-period exclusions, legal edge cases, leave-one-out influence, placebo timing, and staggered-adoption summaries. Heterogeneity and policy-feature analyses are exploratory.

Results

Adoption was concentrated in higher-risk states

Before estimating mortality models, we compared clean-adopter and verified non-adopter states on baseline state characteristics (Fig. 1). Adopter states had higher Republican two-party presidential vote share (0.579 vs. 0.421), higher estimated household gun ownership (0.476 vs. 0.322), greater rurality (0.679 vs. 0.433), and higher baseline firearm-suicide rates (8.849 vs. 5.435 deaths per 100,000). Figure 1 shows that adopting states were concentrated toward higher baseline firearm-suicide rates and higher Republican two-party presidential vote share. This pattern shows that adoption was politically patterned and correlated with baseline firearm-suicide risk rather than distributed across comparable states.

Policy adoption is patterned by baseline risk and politics



Point size is proportional to baseline household firearm ownership. State-level baseline measures describe selection, not treatment effects.

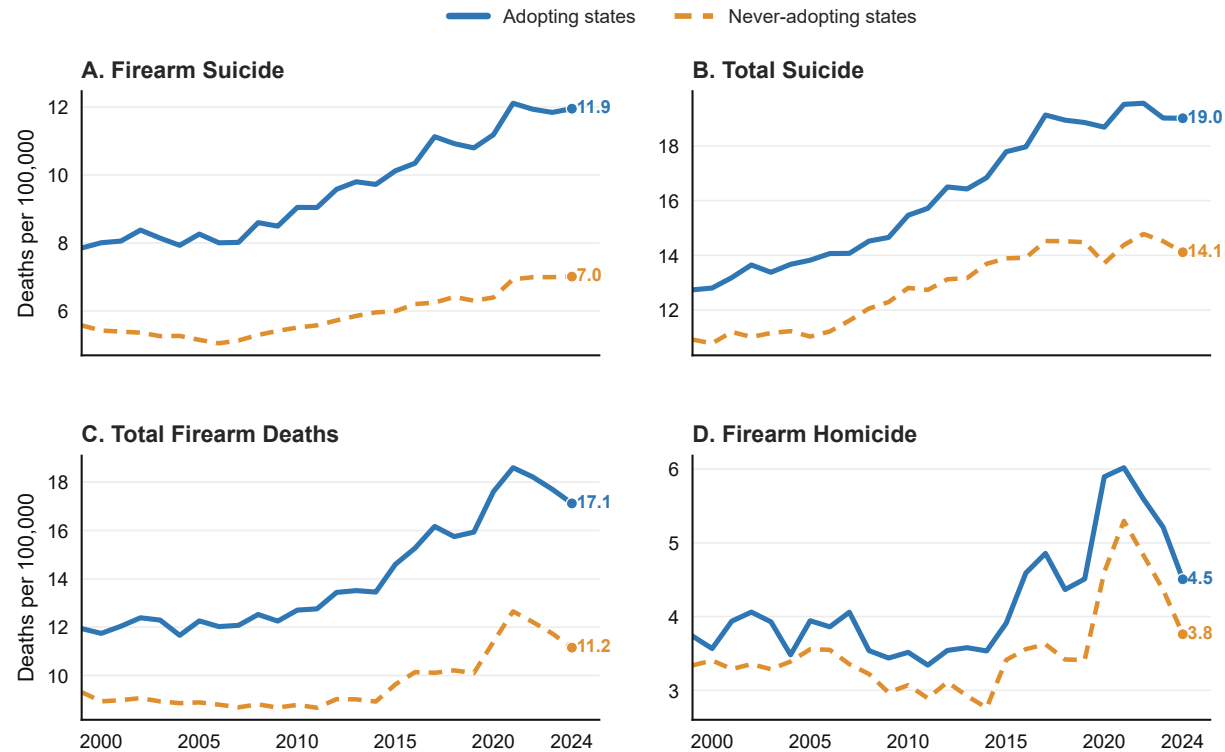
Figure 1: **Selection into permitless carry adoption.** States that adopted permitless carry were concentrated toward higher baseline firearm-suicide rates and higher Republican two-party presidential vote share. Point size reflects estimated household gun ownership.

The formal adoption logit was not stable because the small state sample and strong political sorting produced perfect-separation and convergence problems. We therefore use the descriptive comparisons as evidence of selection rather than interpreting unstable logit coefficients. This pattern frames the central empirical limitation: adopting and non-adopting states are not exchangeable without modeling assumptions.

Descriptive trends differed by outcome

Descriptive trends separated suicide and homicide before regression adjustment (Fig. 2). Firearm-suicide and total-suicide rates were higher in adopter states, while homicide trends were noisier and did not visually track the same post-adoption pattern. These trends are not causal evidence because states differed before adoption, but they motivate treating firearm suicide and firearm homicide as distinct endpoints.

Outcome trends by permitless carry adoption status



Annual state means, 1999-2024. Groups are defined by whether a state ever adopts permitless carry during the panel.

Figure 2: **Outcome trends by permitless-carry adoption status.** Descriptive trends motivate separate suicide and homicide analyses. They are not interpreted as causal estimates because adopter and non-adopter states differ structurally and politically before adoption.

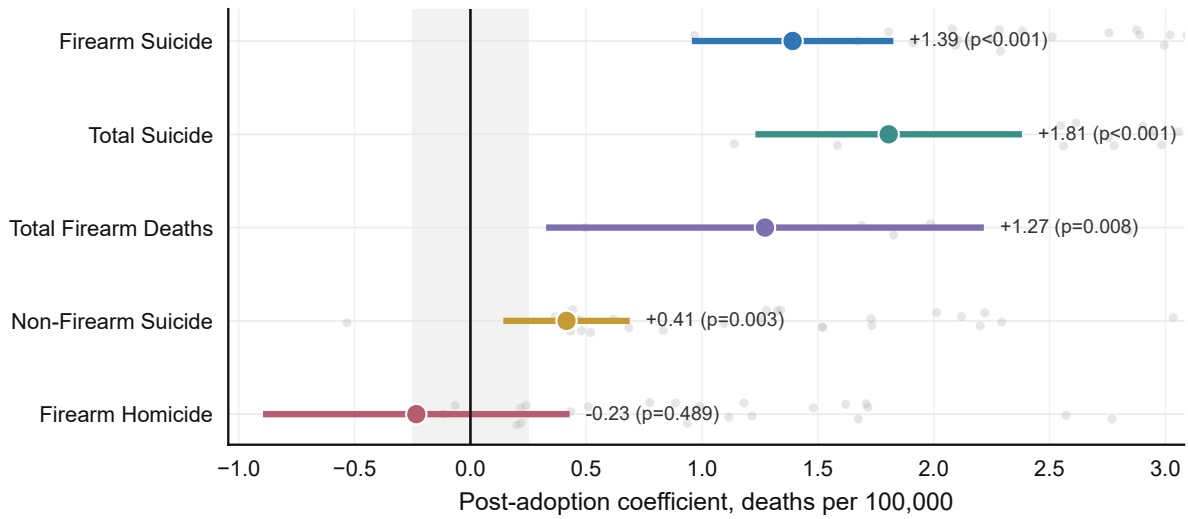
The primary association was concentrated in firearm suicide

The baseline fixed-effects model estimated a positive post-adoption association for firearm suicide (Table 2; Fig. 3). Permitless carry adoption was associated with a 1.391 deaths per 100,000 increase in firearm suicide rates (SE 0.222; $p < 0.001$). Total suicide also increased by 1.805 deaths per 100,000 (SE 0.294; $p < 0.001$), and total firearm deaths increased by 1.271 deaths per 100,000 (SE 0.482; $p = 0.008$). Non-firearm suicide was positive but smaller (0.415; SE 0.139; $p = 0.003$). Firearm homicide did not show a comparable association (-0.234; SE 0.338; $p = 0.489$). This baseline pattern shows that permitless carry adoption was associated most clearly with suicide-related mortality, not with firearm homicide.

Table 2: **Baseline fixed-effects mortality estimates.**

Outcome	Coef.	SE	<i>p</i>	N
Firearm suicide	1.391	0.222	< 0.001	1222
Non-firearm suicide	0.415	0.139	0.003	1222
Total suicide	1.805	0.294	< 0.001	1222
Firearm homicide	-0.234	0.338	0.489	1142
Total firearm deaths	1.271	0.482	0.008	1222

Adjusted difference-in-differences estimates



Colored estimates are adjusted TWFE coefficients with ± 1.96 SE intervals; pale points show unadjusted adopter-state pre/post changes.

Figure 3: **Baseline fixed-effects coefficient forest.** Models include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. Firearm suicide, total suicide, and total firearm deaths are positive in the baseline specification; firearm homicide is not.

Age-adjusted outcome rates preserved the main pattern

Because suicide mortality differs by age structure across states, we also estimated the baseline models using CDC WONDER age-adjusted rates (Table 3). The firearm-suicide estimate was nearly unchanged after age adjustment (1.407 vs. 1.391 deaths per 100,000). Total suicide and total firearm deaths remained positive, while firearm homicide remained statistically weak and below zero. This check supports the interpretation that the firearm-suicide result is not simply an artifact of different state age composition.

Table 3: **Age-adjusted outcome-rate sensitivity.**

Outcome	Coef.	SE	p	N
Firearm suicide, age-adjusted	1.407	0.228	< 0.001	1220
Total suicide, age-adjusted	1.914	0.327	< 0.001	1222
Total firearm deaths, age-adjusted	1.230	0.510	0.016	1222
Firearm homicide, age-adjusted	-0.230	0.426	0.590	1037

Table 4 places the firearm-suicide estimate beside the main sensitivity checks. The association is positive across many specifications, including population weighting, pre-COVID restriction, COVID-period exclusion, and external firearm-law covariates. State-specific linear trends attenuate the estimate to 0.249 ($p = 0.079$), and event-time fixed effects show pre-adoption signal. These rows are the main caution for causal interpretation.

Table 4: **Firearm-suicide diagnostic spine.**

Diagnostic row	Estimate	Diag. p	Interpretation
Baseline unweighted fixed effects	1.391	< 0.001	Clean-adoption state-level association
Population-weighted fixed effects	0.919	< 0.001	Resident-weighted association
Pre-COVID only	1.298	< 0.001	Drops post-2019 years
COVID-excluded	1.356	< 0.001	Drops 2020–2021
External firearm-law covariates	1.096	< 0.001	Policy-environment sensitivity
Leave-one-adopter-out	1.265–1.481	–	No sign changes across adopter exclusions
Never-treated placebo timing	1.391 > 0.338	–	Observed estimate exceeds placebo timing distribution
State-specific linear trends	0.249	0.079	Central attenuation caveat
Cohort ATT windows	0.601/0.577/0.641	–	2-, 3-, and 5-year windows
Not-yet-treated dynamic ATT	0.387	–	Positive post-adoption mean
Event-time fixed effects	0.651	0.0000001	Positive post mean; diagnostic p is the minimum pre-adoption p

Estimates are deaths per 100,000 residents. TWFE rows include state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors unless otherwise noted. Modern staggered-adoption rows are reported as diagnostics because event-time checks flag pre-adoption signal; the event-time p column reports the minimum pre-adoption p rather than a post-adoption effect test.

Firearm-homicide missingness was concentrated in small states

Firearm homicide did not follow the same pattern. The baseline fixed-effects estimate was -0.234 deaths per 100,000 (SE 0.338; $p = 0.489$), with 1,142 state-year observations. Missingness remains relevant: 80 of 1,222 possible clean-primary state-years were missing or suppressed, all in the smallest-population quartile (Table 5). This limitation tempers any strong homicide-null claim, but it does not explain the suicide-homicide contrast.

Table 5: **Firearm-homicide missingness audit.**

Group	Possible	Observed	Missing	Missing pct.
All state-years	1222	1142	80	6.5%
Adopter-state state-years	676	610	66	9.8%
Non-adopter-state state-years	546	532	14	2.6%
Q1 smallest	312	232	80	25.6%
Q2	312	312	0	0.0%
Q3	286	286	0	0.0%
Q4 largest	312	312	0	0.0%

Missing firearm-homicide rates are concentrated in the smallest-population state-years. No state-year in population quartiles 2–4 is missing the firearm-homicide rate.

Together, the baseline estimates and missingness audit support an outcome-specific interpretation rather than a broad firearm-violence claim.

Change-score windows showed the same outcome contrast

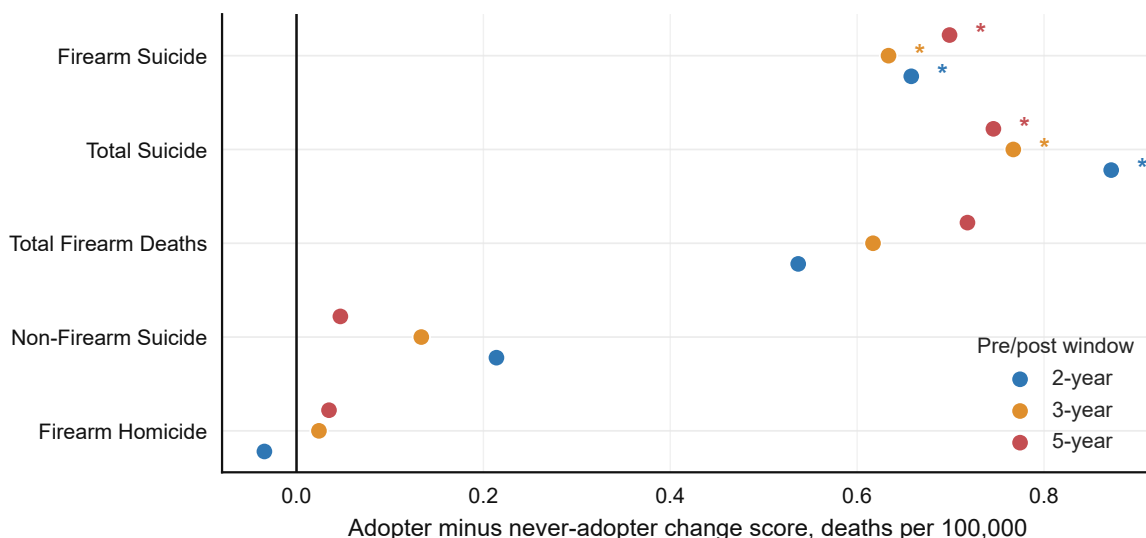
The pre-post change-score design asks a simpler question: within fixed windows around adoption, did adopter states experience larger changes than never-adopter states? Firearm suicide was positive and statistically significant in all three windows. The adopter-minus-never difference was 0.658 deaths per 100,000 in the 2-year window ($p = 0.001$), 0.634 in the 3-year window ($p < 0.001$), and 0.699 in the 5-year window ($p < 0.001$).

Firearm homicide again showed little evidence of a post-adoption increase (Table 6; Fig. 4). The corresponding homicide differences were -0.034 ($p = 0.881$), 0.024 ($p = 0.928$), and 0.035 ($p = 0.899$). Total suicide was positive in all three windows, while total firearm deaths was positive but did not reach $p < 0.05$ in these window checks. Although less adjusted than the fixed-effects model, the change-score comparison shows that the suicide-homicide contrast is not unique to the baseline regression specification.

Table 6: **Pre-post change-score window checks.**

Outcome	2-year diff.	3-year diff.	5-year diff.
Firearm suicide	0.658**	0.634***	0.699***
Non-firearm suicide	0.214	0.134	0.047
Total suicide	0.872**	0.767***	0.746**
Firearm homicide	-0.034	0.024	0.035
Total firearm deaths	0.537	0.617	0.718

Change-score comparisons across robustness windows



Positive values indicate larger pre/post increases in adopting states. Asterisks denote Welch-test $p < 0.05$.

Figure 4: **Change-score robustness across pre-post windows.** Firearm suicide is positive across 2-, 3-, and 5-year windows. Firearm homicide remains statistically weak across the same windows.

Covariate layers attenuated the firearm-suicide estimate

Permitless carry may move alongside other firearm policies. We therefore added covariates for permit-to-purchase laws, waiting periods, universal background checks, ERPO/red-flag laws, safe-storage laws, stand-your-ground laws, and dealer licensing using the Tufts database [14]. The firearm-suicide estimate decreased from 1.391 to 1.096 deaths per 100,000 but remained statistically significant ($p < 0.001$). Total suicide and total firearm deaths also remained positive; firearm homicide remained statistically weak.

Suicide-related social environments may also differ across adopting and non-adopting states. Health-access covariates reduced the firearm-suicide estimate to 0.707 ($p < 0.001$), overdose covariates reduced it to 0.492 ($p = 0.060$), and demographic-poverty covariates reduced it to 0.695 ($p < 0.001$). The estimate stayed positive in these rows, but the magnitude changed.

The strongest attenuation came from specifications that included mental-health-provider access (Table 7). The HRSA mental-health-provider window is short, leaving only 141 state-year observations for the mental-health and full Phase 3B2 models. In that shorter window, firearm suicide remained positive but was no longer statistically significant (0.116, $p = 0.744$ for mental-health access; 0.176, $p = 0.633$ for the combined Phase 3B2 specification). These rows mark a boundary for the full-panel association rather than replacing it.

Table 7: **Firearm-suicide estimate across covariate-layer sensitivities.**

Specification	Coef.	p	N
Baseline fixed effects	1.391	< 0.001	1222
External firearm-law covariates	1.096	< 0.001	1222
Health-access covariates	0.707	< 0.001	752
Overdose covariates	0.492	0.060	282
Demographic-poverty covariates	0.695	< 0.001	940
Mental-health access covariates	0.116	0.744	141
Full Phase 3B2 covariates	0.176	0.633	141

Staggered-adoption analyses raised pretrend concerns

Because adoption is staggered across states, we next examined cohort-window and event-time estimates (Table 8). Cohort-window ATT estimates for firearm suicide were positive across the 2-, 3-, and 5-year windows: 0.601, 0.577, and 0.641 deaths per 100,000. The not-yet-treated dynamic ATT summary also had a positive post-adoption mean for firearm suicide (0.387), while firearm homicide had a negative post-adoption mean (-0.157). These estimates were smaller than the baseline TWFE estimate, so they support the direction of the association more than its exact magnitude.

Table 8: **Staggered-adoption estimate summary.**

Outcome	ATT w2	ATT w3	ATT w5	Not-yet post	Interpretation
Firearm suicide	0.601	0.577	0.641	0.387	Positive, smaller than TWFE
Non-firearm suicide	0.270	0.239	0.241	0.077	Positive, smaller than TWFE
Total suicide	0.871	0.816	0.882	0.464	Positive, pretrend flag
Firearm homicide	-0.004	-0.107	-0.106	-0.157	No positive estimate
Total firearm deaths	0.382	0.435	0.493	0.200	Positive, pretrend flag

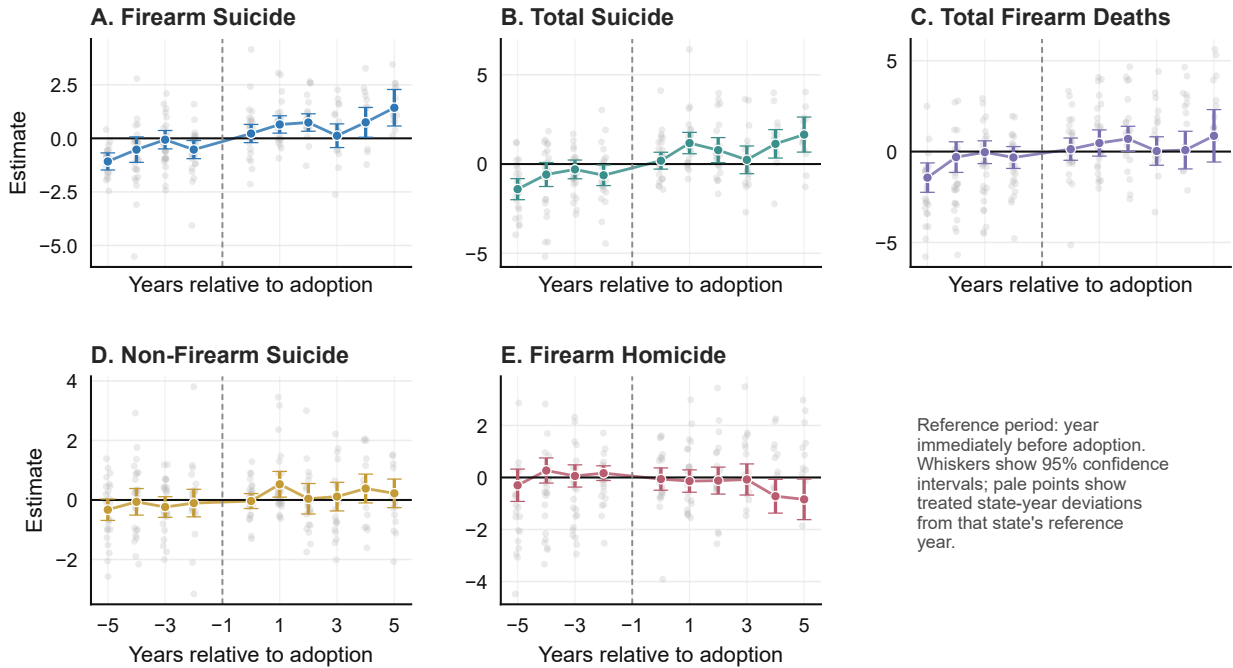
The event-time estimates are more cautionary (Fig. 5). Firearm suicide had positive post-adoption coefficients, but it also had statistically significant pre-adoption coefficients. Total suicide and total firearm deaths showed similar pretrend patterns. Firearm homicide had no positive post-adoption pattern.

Robustness checks identified trend sensitivity

The firearm-suicide estimate remained positive under COVID-period exclusion (1.356; $p < 0.001$), pre-COVID restriction (1.298; $p < 0.001$), population weighting (0.919; $p < 0.001$), and external firearm-law covariates (1.096; $p < 0.001$). Leave-one-adopter-out estimates ranged from 1.265 to 1.481, and the observed firearm-suicide coefficient exceeded the 95th percentile of the never-treated placebo-timing distribution.

State-specific linear trends were the main exception. With state trends, the firearm-suicide estimate attenuated to 0.249 and was not statistically significant at the 0.05 level ($p = 0.079$). Total firearm deaths also attenuated strongly (0.070; $p = 0.806$). Total suicide remained positive (0.372; $p = 0.025$).

Event-study estimates around adoption



Dynamic coefficients are plotted relative to the year before permitless carry adoption; background points show the underlying treated-state event-time data.

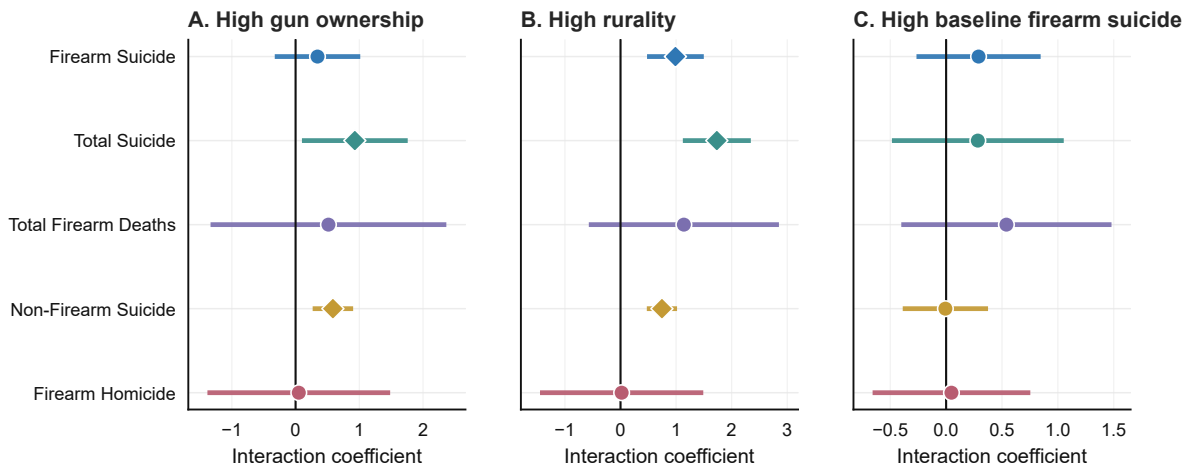
Figure 5: **Event-time diagnostic estimates.** Event-time models show post-adoption suicide-related increases but also pre-adoption signals for firearm suicide, total suicide, and total firearm deaths.

Legal edge-case sensitivity did not change the broad pattern. When Arkansas was coded as treated in 2021 or 2023, firearm suicide remained positive and significant (1.361 and 1.379), total suicide remained positive and significant (1.777 and 1.785), and firearm homicide remained statistically weak (-0.205 and -0.250). Reintroducing Mississippi alone produced a firearm-suicide estimate of 1.300, and the combined Mississippi-plus-Arkansas scenarios remained positive and significant. The main outcome contrast is therefore not driven by these edge-case coding decisions.

Heterogeneity patterns were strongest for rurality

The clearest heterogeneity pattern was rurality (Fig. 6). Firearm suicide had a positive high-rurality interaction (0.988; $p < 0.001$), total suicide also had a positive high-rurality interaction (1.733; $p < 0.001$), and non-firearm suicide had positive interactions for high rurality (0.745; $p < 0.001$) and high estimated gun ownership (0.586; $p < 0.001$). Total firearm-death and firearm-homicide heterogeneity estimates were not statistically strong.

Heterogeneity in post-adoption associations



Interaction terms estimate whether associations are larger in states above each baseline median; diamonds mark $p < 0.05$.

Figure 6: **Exploratory heterogeneity checks.** Rurality is the clearest recurring heterogeneity dimension for firearm suicide, total suicide, and total firearm deaths. These models are exploratory because subgroup indicators are coarse and adoption is politically patterned.

The data do not directly observe carrying frequency, firearm acquisition, firearm storage, crisis timing, impulsive crises, or household firearm access. Plausible pathways include reduced pre-carry screening, reduced delay, weaker training requirements, and normalization of firearm availability, but the state-year panel cannot distinguish among them. The mortality pattern is therefore consistent with higher suicide-related risk after adoption, but it does not identify a specific firearm-access mechanism.

The policy-feature models did not identify a stable mechanism. None of the 15 training, permit-background-check, or misdemeanor-screen interaction rows had $p < 0.05$. The permit-background-check interaction could not be estimated because all 26 source-verified adopter rows removed the carry-permit background-check screen or an equivalent carry-permit screen, leaving no source-verified comparison group. Training removal and misdemeanor-screen removal had estimable rows, but the interaction estimates were not statistically strong. These results are best read as descriptive feature checks rather than evidence that one permit component explains the mortality association.

Discussion

Permitless carry is usually argued over as a crime policy. The clearest signal in these data is different. In the 1999–2024 state panel, states that removed concealed-carry permit requirements had larger increases in firearm suicide after adoption. Firearm homicide did not move in the same way.

That contrast matters. Removing a carry permit is not just another version of shall-issue permitting, and it is not only a question about deterrence or public crime. The permit process is one of the few points where a state can delay, document, train, or screen a person before concealed carry in public. Once that step disappears, suicide belongs near the center of the analysis rather than as a secondary outcome behind homicide or violent crime.

The firearm-suicide estimate was positive in the baseline model, the age-adjusted model,

population-weighted models, COVID-period exclusions, external firearm-law models, Arkansas recoding, Mississippi reintroduction, leave-one-adopter-out checks, and cohort-window ATT summaries. The size was not identical across methods. The baseline estimate was 1.391 deaths per 100,000, while the cohort-window ATT estimates were closer to 0.6. The cleaner interpretation is that the firearm-suicide association keeps reappearing, but its magnitude depends on the comparison being made.

The main caution is that adopting states were not starting from the same place as non-adopting states. They already had higher firearm-suicide rates, higher estimated gun ownership, greater rurality, and higher Republican presidential vote share. State-specific trends reduced the firearm-suicide estimate, and event-time models showed movement before adoption. The post-adoption increase is present in the panel, but it sits on top of state trajectories that were already different before the laws changed.

The mechanism is still unresolved. If the only pathway were immediate access to a firearm during a suicidal crisis, one might expect firearm suicide to rise while non-firearm suicide stayed flat or fell. Instead, non-firearm suicide was also positive in the baseline model, although smaller and less stable. The state-year data cannot tell us who carried, who bought a firearm, how firearms were stored, or whether a death involved someone affected directly by the permitless-carry law.

The homicide results point in a different direction. Firearm homicide was not positive in the baseline model, the age-adjusted model, the change-score comparisons, or the staggered-adoption summaries. Missing or suppressed homicide rates in small-population state-years mean we should be careful about claiming no homicide effect. But in the available data, homicide does not show the same post-adoption pattern as firearm suicide.

The legal audit is part of the evidence. Arkansas, Mississippi, and Vermont show why the treatment cannot be coded as a simple constitutional-carry label. A state can be baseline permitless, partially permitless, newly clarified by statute, or ambiguous during part of the panel. Separating those cases keeps the analysis attached to the actual legal change.

The feature analysis did not identify a stable mechanism. Training removal, background-check removal, and misdemeanor-screen removal are plausible pathways, but feature groups are small or nearly universal among clean adopters. The absence of statistically significant feature interactions is therefore not strong evidence against those pathways.

These findings place permitless carry closer to the firearm-suicide and lethal-means literature than the older right-to-carry crime debate alone. State firearm laws associated with lower homicide are not always the same as those associated with lower suicide [21, 22]. Recent suicide-specific evidence links handgun permits, waiting periods, and concealed-carry permit requirements to lower firearm suicide rates [7]. Public carry may still matter for crime, but that was not where this panel showed its clearest mortality signal.

Future work should treat firearm suicide as a primary endpoint, audit legal timing state by state, and connect law changes to behavior: carrying, acquisition, storage, crisis timing, and decedent-level firearm access. Another state-year coefficient alone will not answer those questions.

Additional Methods Details

All baseline fixed-effects models used ordinary least squares with state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors. Robustness specifications excluded 2020–2021, restricted the panel to pre-2020 years, applied population weights, added state-specific linear trends, or added external firearm-law covariates. Covariate-layer sensitivity models added firearm-law domains, uninsured rates, overdose mortality, age structure, race and ethnicity

shares, poverty rates, and mental-health-provider access as described in the repository output tables. The unweighted estimates target the average state-level association; the population-weighted estimates target the average resident-level association.

Change-score analyses computed, for each adopter state, the difference between the average post-adoption rate and the average pre-adoption rate in 2-, 3-, and 5-year windows. Never-adopter states were assigned comparable window summaries and compared with Welch tests. Staggered-adoption checks included cohort-window ATT estimates, not-yet-treated dynamic ATT estimates, and never-treated-control event-time fixed-effect estimates. Age-adjusted outcome models used CDC WONDER age-adjusted rates as alternate outcomes and retained the same state and year fixed effects, unemployment, per-capita income, and state-clustered standard errors as the crude-rate baseline model. These checks were designed as sensitivity analyses because several outcomes showed pre-adoption event-time signals.

The outcome order was fixed before drafting the paper but was not preregistered. Firearm suicide is the primary outcome. Total suicide, non-firearm suicide, firearm homicide, and total firearm deaths are secondary outcome-family checks. Firearm-law covariates, contextual covariate layers, age-adjusted outcome rates, staggered-adoption estimators, Arkansas recoding, leave-one-out influence, and placebo timing are sensitivity analyses. Rurality, gun ownership, baseline-risk heterogeneity, and policy-feature interactions are exploratory.

Data and Code Availability

Processed analysis outputs, source-audited policy coding files, figure-generation scripts, and the manuscript-ready result tables are maintained in the project repository. Raw CDC mortality files and firearm-law source files are governed by their originating agencies and are documented in the repository source appendix.

Author Contributions

B.K. and Y.W. assembled the policy audit, implemented the mortality panel analyses, generated figures and tables, interpreted the results, and drafted the manuscript.

Competing Interests

The authors declare no competing interests.

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